

# Youssef Kabbaj

MSc Applied Physics — machine learning for molecular and physical systems

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EPFL graduate: facilitated Swiss work authorisation  
Available from April 2027

## Profile

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MSc student in Applied Physics at EPFL (graduating March 2027), working at the interface of **statistical physics** and **generative machine learning**. Currently developing **discrete diffusion models** for conditioned **protein sequence generation**; previously built transformer-parametrised **Potts models** (attention DCA) for protein families, and a deep-learning pipeline for ultrasound motion estimation and object localization in industry. Comfortable across the stack: stochastic processes and sampling theory, PyTorch training on GPU, and molecular simulation (GROMACS, CP2K, QM/MM).

## Education

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| EPFL — Master of Science in Applied Physics<br>GPA 5.68 / 6 — expected March 2027  | 09/2024 – present |
| EPFL — Bachelor of Science in Physics<br>GPA 5.69 / 6                              | 09/2021 – 07/2024 |
| EPFL — Cours de Mathématiques Spéciales (CMS)<br>GPA 5.87 / 6                      | 09/2020 – 07/2021 |
| Groupe Scolaire Atlas, Morocco — Baccalauréat, Mathematical Sciences<br>18.94 / 20 | 07/2020           |

## Technical skills

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- **Statistical physics & stochastic modelling** — Langevin and Fokker–Planck equations, stochastic differential equations, Monte Carlo and MCMC sampling, maximum-entropy and Potts models, free-energy landscapes.
- **Generative modelling** — diffusion and score-based models, flow matching, autoregressive models, attention and transformers, energy-based models; training by maximum and pseudo-likelihood.
- **Machine learning** — supervised and reinforcement learning, synthetic data generation with ground truth, metric design for unlabelled data, model evaluation and failure analysis.
- **Molecular simulation** — set up and ran QM/MM and molecular-dynamics workflows in GROMACS, CP2K and MiMiC during a semester project; structure visualisation in PyMOL.
- **Chemistry foundations** — bond types, intermolecular interactions, binding and affinity energies: enough to work alongside chemists and read structural data critically.
- **Programming** — Python (PyTorch, NumPy, scikit-learn), C++ , MATLAB, Git, Linux; basic experience running jobs on HPC clusters.

## Research and industry experience

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**Laboratory of Statistical Biophysics, EPFL** 11/2026 – 03/2027  
*Master thesis (from November 2026):* Diffusion models for conditioned protein sequence generation and their dynamics

- Implement a discrete diffusion model for protein sequence generation.
- Study the dynamics of generation: speciation time and mode collapse.
- Condition generation on protein family and function.

**Bracco Suisse S.A.** 02/2026 – 08/2026  
*Industry internship:* deep-learning pipeline for contrast-enhanced ultrasound image processing

- Built and trained two deep-learning models on ultrasound sequences: one for **optical flow estimation**, one for **localisation**.
- Wrote a dedicated synthetic data generator for each model, designed to reproduce the appearance and statistics of real acquisitions, so that both could be measured against known ground truth.
- On synthetic data, the optical-flow model reached a motion-correction error of roughly 5%, and the localisation model a precision and accuracy both above 95%.

- Designed reference-free metrics to quantify performance on real, unlabelled clinical data where no ground truth exists, and fed the observed failure modes back into the data generators.

#### Laboratory of Computational Chemistry and Biochemistry, EPFL

09/2025 – 11/2025

*Semester project: QM/MM simulation of acetone — GROMACS, CP2K, MiMiC*

- Ran QM/MM simulations of acetone in MiMiC and measured how different placements of the boundary between the quantum and classical regions change the computed energies and vibrational frequencies.

#### Laboratory of Statistical Biophysics, EPFL

02/2025 – 07/2025

*Summer internship and semester project: Biased generative protein models based on attention DCA*

- Fitted a Potts model to the bacterial J-domain alignment ( $L = 63$  sites,  $q = 21$  states, 36,713 sequences), with the couplings parametrised by a factored self-attention layer and trained by reweighted pseudo-likelihood maximisation.
- Sampled new sequences with a Gibbs-like conditional sampler at controlled inverse temperature: generated sequences reproduce the natural one-site frequencies at  $r = 0.99$  and the connected two-site correlations at  $r = 0.87$ , against a train-versus-test ceiling of 0.94.
- Showed the samples are genuinely novel and as diverse as natural ones (mean Hamming distance 0.42 to the nearest training sequence; mean pairwise distance 0.68 versus 0.60), and diagnosed the one clear defect: a sub-family carrying 14.1% of the natural weight but only 5.1% of the samples.
- Found that the supplied train/test files shared 79% of their sequences verbatim; rebuilt a sequence-disjoint split by clustering at 80% identity and re-evaluated everything on it. Code: [github.com/yk985/attention\\_dca\\_jdomains](https://github.com/yk985/attention_dca_jdomains)

#### Swiss Plasma Center, EPFL

07/2024 – 01/2025

*Semester project: Investigation of the Alfvén wave continuum in tokamaks; preceded by a Summer-in-the-Lab research internship*

- Computed the Alfvén continuum spectrum of tokamak equilibria and analysed the frequency gaps opened in it by the coupling between shear Alfvén and acoustic waves.

#### EPFL

09/2022 – present

*Teaching assistant in analysis and electromagnetism (12 h/week)*

### Selected personal projects

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Code and documentation: [github.com/yk985](https://github.com/yk985)

- **Generative models for protein sequences** — built a hierarchical Potts model as a synthetic ground truth with known statistics, then benchmarked autoregressive and diffusion generators against the exact target distribution. [github.com/yk985/hierarchical-Potts\\_models\\_learning](https://github.com/yk985/hierarchical-Potts_models_learning)
- **Drift-field inference from stochastic trajectories** — recovering force fields from simulated overdamped walkers, comparing five approaches from direct estimation to an amortised CNN that infers the field in a single forward pass. [github.com/yk985/drift\\_field\\_inference](https://github.com/yk985/drift_field_inference)
- **Reinforcement learning for microswimmer navigation** — RL policies for a swimmer steering through a noisy low-Reynolds-number environment, benchmarked against the exact dynamic-programming optimum so that the policy gap is measured rather than assumed. [github.com/yk985/microswimmer-rl](https://github.com/yk985/microswimmer-rl)

### Languages

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Arabic — mother tongue · French — C1 · English — C1

Willing to learn German for a role in German-speaking Switzerland.

### Interests

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Basketball and sport in general; contemplation and the larger questions about life. I will bring home-made tiramisu to the office.